E-COMMERCE BIG DATA ANALYTICS PROJECT

Technical Report

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Project: FinalProjectBD
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1. Introduction

This project presents a complete e-commerce big data analytics workflow. The work covers synthetic data generation, data storage in MongoDB and HBase, data processing with Apache Spark, integrated analytics, and data visualization. The purpose was to build a practical multi-model analytics system and demonstrate how different big data tools can work together in one project.

2. Project Objectives

- Generate a realistic e-commerce dataset.
- Store and analyze data in MongoDB.
- Store session and browsing metrics in HBase.
- Use Apache Spark for cleaning and large-scale analytics.
- Integrate outputs from MongoDB and HBase.
- Create visualizations and final reporting deliverables.

3. Dataset Generation

The dataset was generated using a custom Python generator. The generator created users, categories, products, transactions, and browsing sessions.

Dataset	Count
Users	10,000
Products	5,000
Categories	25
Transactions	500,000
Sessions	2,000,000

The generated JSON files were validated successfully before loading into the storage systems.

4. MongoDB Implementation

MongoDB was used to store the main project data. The collections created in the ecommerce_db database were users, products, categories, transactions, and sessions.

Collection	Documents
categories	25
products	5,000
sessions	2,000,000
transactions	500,000
users	10,000

Key MongoDB findings:

- Total transactions: 500,000
- Overall transaction value: 466,576,163.45

- Completed transactions: 317,055
- Overall conversion rate: 20.29%
- Traffic sources were almost evenly distributed across social, affiliate, email, direct, and search_engine.
- Device conversion rates were also similar across tablet, desktop, and mobile.

5. HBase Implementation

HBase was used to store browsing-oriented data in a column-family model. Because the full sessions dataset was very large, a sample of 5,000 sessions from sessions_0.json was loaded into HBase for this stage of the project.

Two HBase tables were created:

- fp_user_sessions: stores session details, page views, and cart data.
- fp_product_metrics: stores product-date browsing metrics.

HBase loading results:

- fp_user_sessions rows: 5,000
- fp_product_metrics rows: 16,804

The HBase row key for sessions used user_id, reverse timestamp logic, and session_id. This design supports user-based and recent-session access patterns.

6. Apache Spark Processing and Analytics

Apache Spark was used for batch analytics. The transactions dataset was converted to NDJSON format and processed with PySpark.

Spark cleaning results:

- Original transactions: 500,000
- Invalid transactions: 0
- Duplicate rows removed: 0
- Clean transactions: 500,000

Spark SQL query highlights:

- Completed status had the highest number of transactions and highest transaction value.
- Monthly transaction activity was observed across August, September, and October 2026.
- Customer transactions were also summarized by registration month.

Monthly transaction trends:

Month	Transactions	Unique Customers	Transaction Value
2026-08	169,268	10,000	157,886,330.43
2026-09	166,157	10,000	154,991,898.17
2026-10	164,575	10,000	153,697,934.85

Cohort analysis:

- Registration cohorts were created from February 2026 to August 2026.
- The observed activity rate in the generated dataset was 100% for the available active months.
- Cohort results were saved successfully for reporting purposes.

7. Integrated Analytics

The integrated analysis combined MongoDB product information, MongoDB product sales metrics, and HBase browsing metrics. The integration was performed using product_id as the common key.

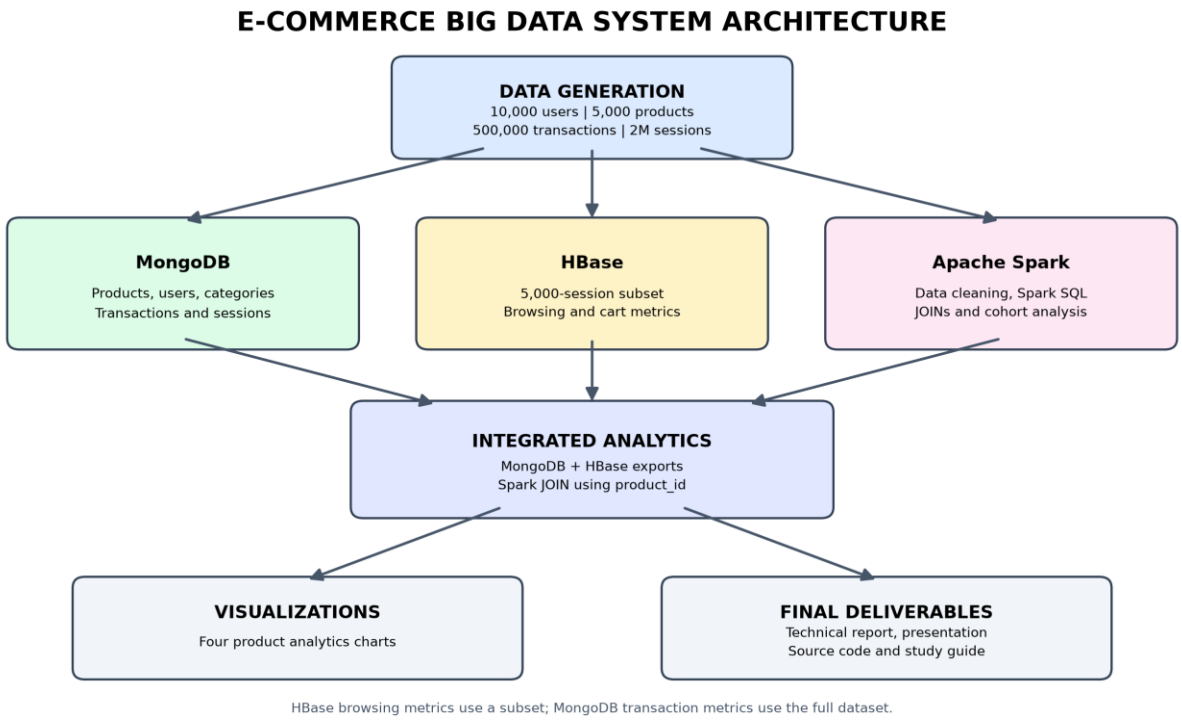
- MongoDB products exported: 5,000
- MongoDB products with transaction items: 4,766
- HBase product-date records: 16,804
- Integrated products: 5,000

Examples of integrated insights:

- Top products were identified by browsing views.
- Top products were identified by completed transaction quantity.
- Browsing views and completed quantities were compared visually.

8. System Architecture

The following architecture diagram summarizes the full project workflow from dataset generation to storage, processing, integration, visualization, and reporting.



9. Visualizations

Four charts were created to summarize the main integrated analytics outputs.

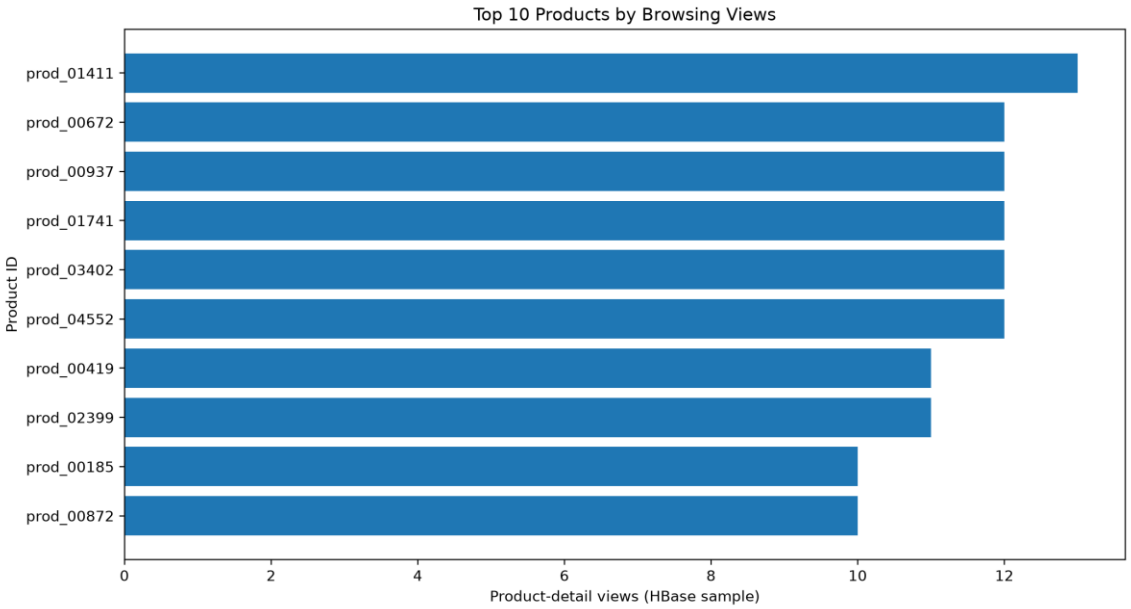


Figure 1. Top products by browsing views.

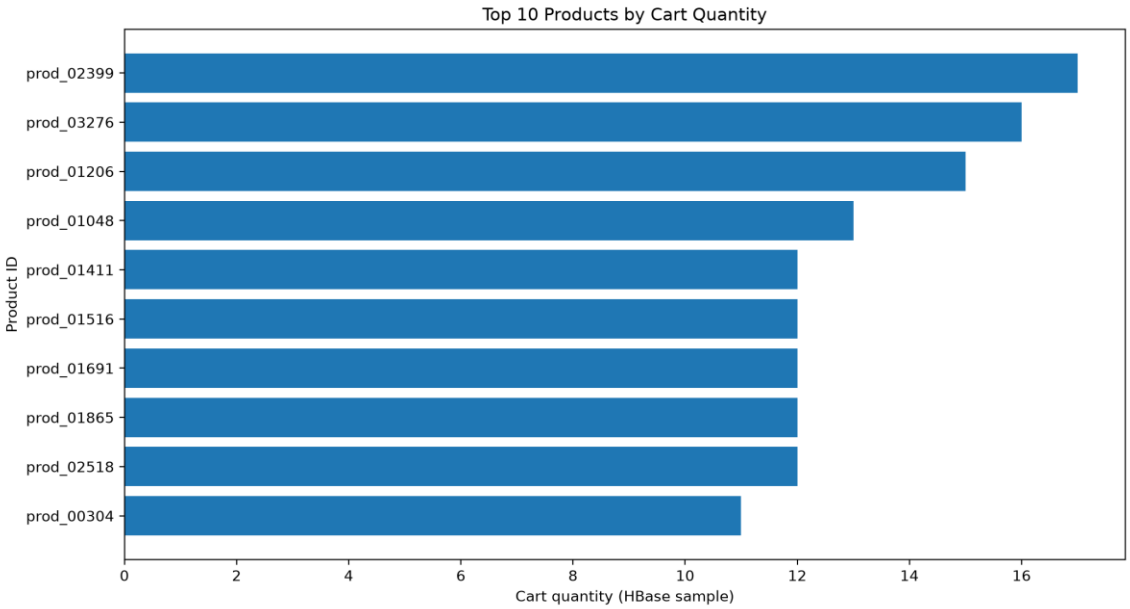


Figure 2. Top products by cart quantity.

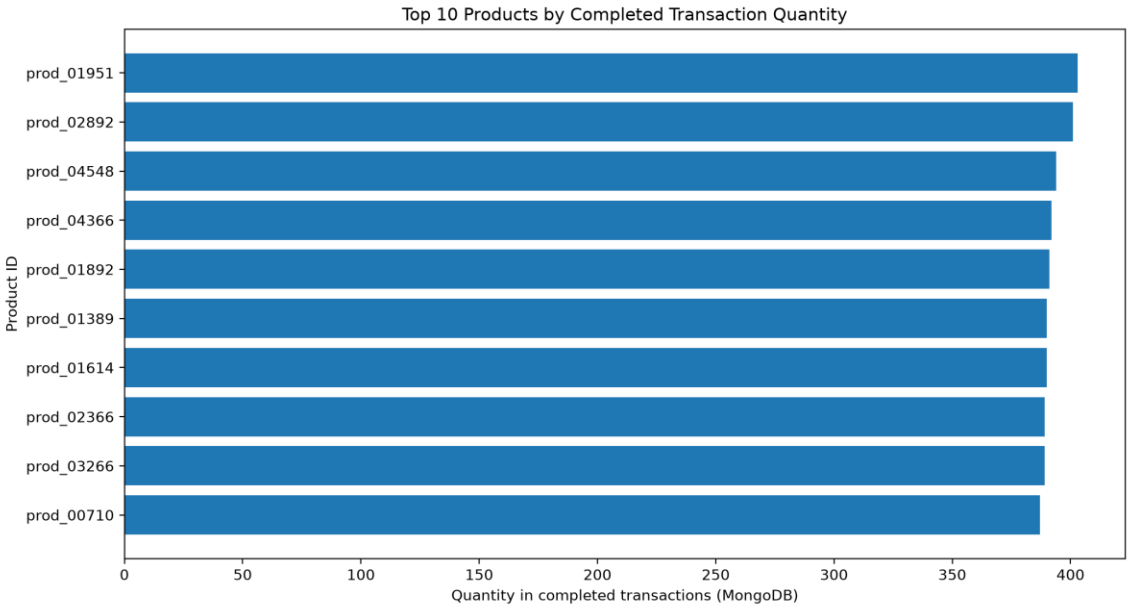


Figure 3. Top products by completed transaction quantity.

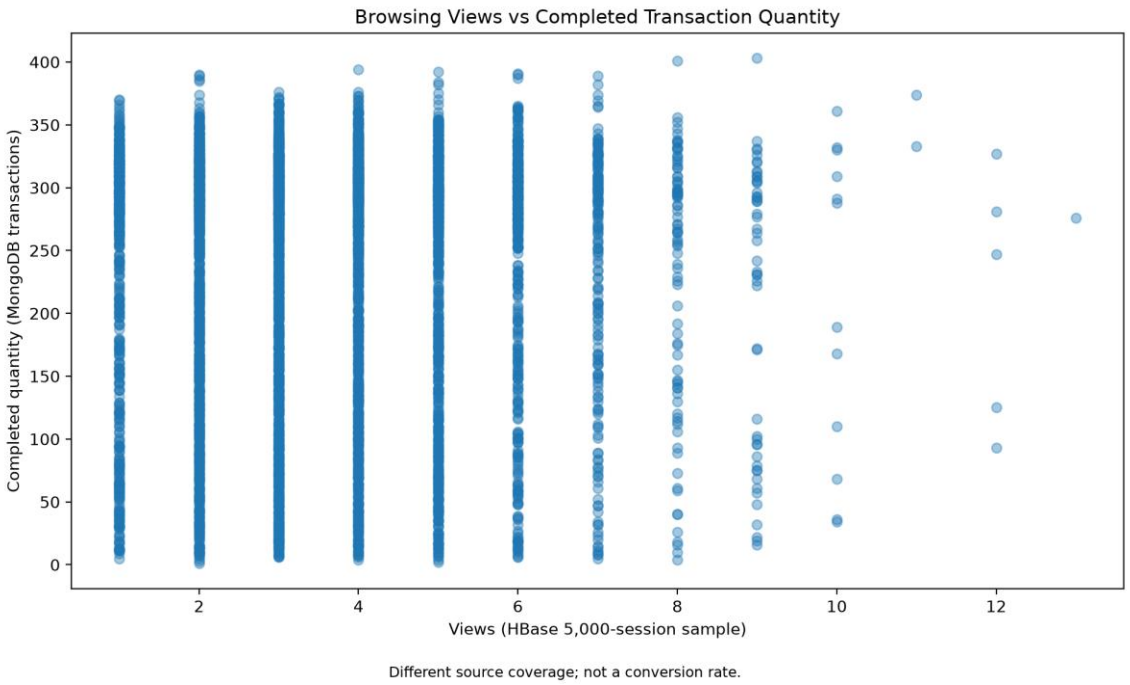


Figure 4. Browsing views versus completed transaction quantity.

10. Challenges Encountered

- MongoDB was not initially installed and had to be configured.
- The original HBase container failed because of the thrift startup issue.
- A recovery container had to be created to run HBase without the problematic thrift startup.
- Spark initially failed because of environment and Java compatibility issues.
- Large data volumes required careful step-by-step validation.

11. Conclusion

This project successfully demonstrated a full big data analytics pipeline for e-commerce data. MongoDB handled document-based storage, HBase supported column-family browsing analytics, and Spark enabled scalable processing and reporting. The final outputs included integrated analysis, visual charts, and system documentation. The project also provided practical experience in debugging, data integration, and multi-tool analytics workflows.